

LESSON 5-4 • SESSION 1 • PRACTICE • FORM A

Area of Regular Polygons

Learning target – I can find the area of a regular polygon by finding the area of one triangle and multiplying. • 6.G.A.1

REMEMBER

- A **regular polygon** has equal sides and equal angles. Lines from the center make **congruent triangles**.
- Each triangle's base is one side. Its height is the **apothem**.
- **Area = $n \times (\frac{1}{2} \times \text{side} \times \text{apothem})$** . Use $\approx$ when the apothem is rounded.

WORKED EXAMPLE

Regular pentagon: sides 10 cm, apothem about 7 cm.

Step 1 5 sides make 5 triangles.**Step 2** One triangle: $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$.**Step 3** Five triangles: $5 \times 35 = 175$.**Answer** Area $\approx 175 \text{ cm}^2$ **WORDS** regular polygon (*polígono regular*) • apothem (*apotema*) • congruent (*congruente*)

PRACTICE

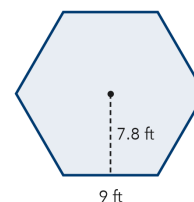
show your work • label units

1. Find the area of one center triangle:
- $\frac{1}{2} \times \text{side} \times \text{apothem}$
- .

a. side 10 cm, apothem 7 cm ____ cm^2 b. side 16 in, apothem 11 in ____ in^2 c. side 7 m, apothem 8 m ____ m^2 d. side 6 ft, apothem 9 ft ____ ft^2

- 2.
- Gazebo Floor.**
- A gazebo floor is a regular hexagon with sides of 9 ft. The apothem is about 7.8 ft.

a. How many center triangles? _____

b. Area of one triangle _____ ft^2 c. Area of the floor _____ ft^2 

- 3.
- Pentagon Sign.**
- A regular pentagon road sign has sides of 16 in and an apothem of about 11 in. Find its area.

Hint A pentagon has 5 center triangles.

Show your work.

Answer _____ in^2

- 4.
- Stepping Stone.**
- A stepping stone is a regular octagon with sides of 7 cm and an apothem of about 8 cm. Find its area.

Hint Find one triangle, then multiply by the number of sides.

Show your work.

Answer _____ cm^2

- 5.
- Flower Bed.**
- A flower bed is a regular hexagon with sides of 24 ft. Each center triangle has a height of about 21 ft. Find the area of the flower bed.

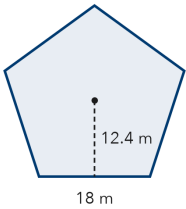
Show your work.

Answer _____ ft^2

6. **Pentagon Patio.** A patio is a regular pentagon. Decompose it into center triangles and find its area.

Hint All the triangles share the same base and height.

Show your work.



Answer _____ m²

7. **Jamal's Tile.** A tile is a regular hexagon with sides of 10 cm and an apothem of about 9 cm. Jamal says the tile's area is 45 cm². Is Jamal right? Explain.

Hint How many triangles make up the whole tile?

Show your work.

True area _____

8. **Tile Design.** A floor uses regular hexagon tiles with sides of 8 cm and an apothem of 6.9 cm.

- a. Find the area of one tile. _____ cm²
b. Find the area covered by 25 tiles. _____ cm²

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Lena's Work.** A regular pentagon has sides of 22 in and an apothem of 15 in. Lena wrote: "A = 22 × 15 × 5 = 1,650 in²." Explain the mistake and find the correct area.

TEST PRACTICE

choose the best answer

10. A regular hexagon splits into how many congruent center triangles?

A 4 **B** 5 **C** 6 **D** 8

11. One center triangle of a regular pentagon has an area of 20 cm². The pentagon's area is –

A 25 cm² **B** 40 cm² **C** 100 cm² **D** 120 cm²

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

440 O	45 Q	1,512 Y	1,116 Z	558 G	224 L	4,140 N	210.6 P	270 O	880 X
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#2c	#3	#4	#5	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____